

Maize — Crop Production Notes

Introduction and Local Names

The scientific name of maize is *Zea mays* L., and it belongs to the family Poaceae. In Urdu, maize is called Makai, and in Punjabi it is known as Maki.

Crop Characteristics

Maize is an annual, short-day crop with a determinate growth habit. It is cross-pollinated, and its seed is referred to as a caryopsis.

Origin

Maize does not have an exact or specific origin, as it is a cross-pollinated crop. Synthetic maize, used mainly for fodder, has low yield potential, while hybrid maize, grown mainly for grain, has high yield potential. Both types originated from primitive maize.

Top Maize Producing Countries

The leading maize-producing countries, along with their production figures, were the USA with 316,165,000 tonnes, China with 177,540,788 tonnes, Brazil with 55,394,800 tonnes, Mexico with 23,301,900 tonnes, and Argentina with 22,676,900 tonnes.

Maize Production in Pakistan (2016-17)

In 2016-17, maize was cultivated on an area of 1.33 million hectares in Pakistan, with a total production of 6.130 million tonnes and a yield of 4.59 tonnes per hectare. Pakistan ranked 23rd in world maize production that year.

Importance

Maize is the third major cereal crop after wheat and rice. It is used to make bread, soups, and similar foods, and its grain is used as flour. It is also used to produce corn oil, popcorn, and energile. The grain contains about 85% carbohydrates and 5–7% oil. Maize is also used as fresh and dried fodder.

Climate Requirements

The minimum temperature required for maize is 10°C, while the optimum temperature range is 16–18°C. The crop is sensitive to very high temperatures, and leaf burn can occur in May–June due to hot winds. Maize prefers high humidity and is more drought resistant at its early growth stages compared with the grain development stage.

Soil Requirements

Maize grows best in medium-textured, well-drained loam soils. It cannot be grown successfully in dry and sandy soils, and it is sensitive to salinity.

Root Development

Maize has an adventitious root system, along with aerial roots that provide additional support to the plant.

Shoot Development

The maize stem is jointed, with pithy internodes that are 2–4 cm in diameter. The stem is thicker at the base than at the upper side, and the base contains the roots. The leaves are arranged alternately along the stem.

Inflorescence

The tassel is the male part of the maize inflorescence and occupies the terminal position on the plant. The cob or ear is the female part and is located in the mid-portion of the stalk. The silk of the cob starts golden and turns brown, eventually drying and falling.

Primary Tillage

Primary tillage for maize involves 3–4 ploughings to break the hard pan, followed by 2–3 plankings and land levelling.

Seed Bed Preparation

Seed bed preparation involves 1–2 ploughings, use of a rotavator, 2 plankings, and the formation of ridges.

Sowing Time

The spring sowing period runs from mid-January to the end of February. The autumn, or seasonal, sowing period runs from the first week of July to 10th August, and early maturing varieties can be sown up to 20th August. The exact sowing time depends on the cultivar and the region or climate.

Seed Rate and Seed Treatment

The recommended seed rate for maize is 8–10 kg per acre. For seed treatment, Imidacloprid 70WS is applied at 7 grams per kg of seed, and Topsin-M 70WP is applied at 2 grams per kg of seed.

Sowing Method

Maize can be sown by broadcasting for fodder purposes, or using an automatic Kharif drill or a dibbler/maize planter, at a depth of 3–4 cm. Row-to-row spacing is 60–75 cm, and in spring planting, ridges are oriented east to west, with sowing done on the south side of the ridges. Plant-to-plant spacing is 15–20 cm, giving a plant population of about 30,000 plants per acre.

Fertilizer (For Soils of Medium Fertility)

For hybrid maize, the recommended fertilizer rate is 110-46-25 kg per acre of NPK. At sowing, 2 bags of DAP and 1 bag of potassium sulphate are applied; this is followed by 1 bag of urea at each of the 0.75 ft, 2 ft, and 3 ft growth stages, and a further 1 bag of urea before flowering.

For synthetic varieties, the recommended fertilizer rate is 95-35-25 kg per acre of NPK. At sowing, 1.5 bags of DAP and 1 bag of potassium sulphate are applied; this is followed by 0.5 bag of urea at the 0.75 ft stage, and 1 bag of urea at each of the 2 ft and 3 ft stages, as well as before flowering.

Irrigation

The first irrigation is given 3–4 days after germination. Subsequent irrigations are applied at 7–10 day intervals in light soils and 10–15 day intervals in heavy soils. During flowering and fertilization, 2 days of dryness can cause a 20% yield reduction, while 1 week of dryness can cause a 50% yield reduction.

For the spring crop, 11–12 irrigations are typically needed, while the Kharif crop requires 10–11 irrigations. The critical growth stages for irrigation are the seedling stage, grand growth stage, tasseling stage, silking stage, soft dough stage, and hard dough stage.

Weed Management Methods

Weed management methods fall into two types. Pre-sowing methods include crop rotation, the Dab method, soil solarization, and the use of herbicides. Post-sowing methods include hoeing, the use of herbicides, and the use of allelochemicals.

Impact of Weeds

Weeds can reduce maize yield by 29–43% or more. Competition between weeds and maize plants is at its maximum 2–5 weeks after crop emergence.

Important Weeds of Maize

Broad leaf weeds include Itsit, Bathu, Krund, Tandla, Lehli, Bakhra, Kulfa, Jangli Palik, Jangli Haloon, Chibber, Dodhak, and Chulai. Narrow leaf weeds, mainly grasses, include Swanki, Madhana, Khabbal, and Baru. Sedges are collectively known as Deela and include *Cyperus rotundus*, *Cyperus iria*, and *Cyperus difformis*.

Pre-Emergence Weed Control

Dual Gold, applied at 800 ml per acre, is used as a pre-emergence herbicide for the control of both narrow and broad leaf weeds.

Post-Emergence Weed Control

For the control of common broad leaf weeds and sedges, Primextra Gold 720SC herbicide is applied at 400–800 ml per acre after the first irrigation, when the field is moist, to effectively control these weeds.

Harvesting

Short duration maize varieties are harvested 90 days after sowing, while long duration varieties are harvested 120 days after sowing. Harvesting readiness is indicated when the cob changes color, the silk turns brown, and the grains become hard. After harvesting, cobs are stored in heaps for 3–4 days.

Maize is harvested when the moisture content of the grains is between 20–25%. At this stage, the cob sheath dries completely, and some drying may also occur on the stalk. Cobs are removed from the standing crop and dried under sunlight until the grain moisture content falls below 15%.

Shelling

Hand-operated or powered maize shellers are preferred over manual shelling for the safe separation of dried maize grains from the cobs. After cleaning, shelled maize grains are stored in jute sacks in a cool, dry, and airy place, either on a cement floor or on wooden planks; storage on earthen floors should be avoided. The storage room temperature is kept at 20–25°C, with relative humidity maintained at 30–50%.

Yield

Maize hybrids are capable of producing more than 120 maunds of grain per acre, while local varieties typically yield 40–50 maunds per acre.

Varieties

Non-hybrid varieties include Sahiwal 2002 and MMRI Yellow.

Hybrid Varieties

Spring hybrids include Pioneer 3333, P 1543, Hicorn 8288, NK 8441, 33 M 15, NK 8711, and 32 F 10. Seasonal hybrids include 3025 Sufaid, 919, Hicorn 984, Hicorn 993, Hicorn 11, NK 6621, and 31 R 88.